

Smart Public Utility Complaint Management System

REVIEW 1 — DETAILED PROJECT DOCUMENTATION

Problem Understanding, Research & Initial DSA Design

Field	Details
Course	Data Structure and Algorithms - II
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Project	Smart Public Utility Complaint Management System
SDG	SDG 11 — Sustainable Cities and Communities
Review	Review 1 — Problem Understanding & Initial Progress
Review Coverage	Unit/Module 1 (Trees) and Unit/Module 2 (Graphs)
Current Progress	25%
Status	Research, problem analysis, DSA applicability study and conceptual design; implementation not yet started

Note: This document is intentionally more detailed than the concise PBL Progress Report – I. It is designed as supporting evidence for Review 1. Completed analytical work is kept separate from proposed/future implementation.

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1. Introduction

Public utility complaint management is a practical smart-city problem because citizens continuously report issues such as damaged roads, non-functional streetlights, water leakage, drainage blockage, garbage accumulation and sanitation problems. When the number of reports becomes large, the challenge is not only storing complaints but deciding which complaint should be handled first, which department or field worker should receive it, and which route can be used to reach the service location efficiently.

Recent research shows that intelligent complaint systems can use machine learning, NLP, image analysis, geolocation, dashboards and predictive analytics. The reviewed literature demonstrates that complaint classification and prioritization are already active research areas. This project therefore does not claim that it is inventing the idea of a smart complaint platform. Instead, it studies the algorithmic layer behind complaint operations from a Data Structures and Algorithms perspective.

The proposed system treats each complaint as a structured record containing information such as category, severity, location, submission time, status and assigned resource. The system can then organize these records using suitable DSA structures, prioritize urgent complaints, model service locations and road connectivity as a graph, and support shortest-path based field routing.

The work completed for Review 1 is primarily analytical and conceptual. It includes understanding the problem, studying four relevant research papers, identifying a defensible project-specific gap, studying DSA-II Units 1 and 2, evaluating the usefulness of the syllabus concepts, and designing the initial complaint-processing workflow. No claim is made that the final complaint engine has already been implemented.

The documentation therefore answers four questions:

- Why is public utility complaint management a meaningful algorithmic problem?
- What do recent research systems already provide?
- Why are particular DSA structures selected, limited or rejected?
- How will the selected structures fit together when implementation begins?

2. Project Domain and Context

2.1 Domain

Domain: Smart public utility complaint management and DSA-based complaint prioritization, resource assignment and location-aware routing.

2.2 Why this domain was selected

The domain naturally contains a high volume of records, changing priorities, multiple service departments, field workers and location relationships. It therefore provides a realistic setting for demonstrating trees, heaps and graphs without forcing unrelated DSA structures into the project.

Entity	Important attributes	Why it matters
Complaint	Complaint ID, category, severity, description, location, timestamp, status, priority	Main unit processed by the system
Department	Department ID, utility type, service area, workload	Determines the responsible service group
Field Worker	Worker ID, department, current location, availability, active jobs	Candidate resource for assignment
Location	Latitude/longitude or graph node, ward/zone	Connects complaints with service areas and road network
Road/Route	Source, destination, distance/travel cost, status	Forms the graph used for route selection
History	Past complaints, category, location, resolution time	Supports hotspot and recurring-issue analysis

3. Problem Statement

The proposed system addresses the problem of efficiently managing a continuously changing stream of public utility complaints when many complaints may arrive at the same time and available service resources are limited.

Existing complaint platforms already provide reporting, classification, routing or tracking features. The academic problem considered here is narrower and algorithmic: how can complaints be organized so that urgent cases are processed first, records can be retrieved efficiently, suitable field resources can be assigned using location information, and routes can be selected over a connected service network?

The problem can be expressed as follows:

```

INPUT: Complaint records + severity + category + location + timestamp
      Department information + worker availability + road network
      ↓
PROCESS: Priority calculation → ordered complaint processing
        Resource selection → location-aware route calculation
        Status updates → historical frequency/hotspot analysis
      ↓
OUTPUT: Prioritized complaints + assigned resources + selected routes
       + service-status information + analytical summaries

```

3.1 Why manual handling becomes difficult

- A large number of complaints may arrive simultaneously, making first-come-first-served handling unsuitable for urgent cases.
- Complaint priority may change as severity, waiting time or service impact changes.
- Searching manually for complaint, worker or department information becomes slower as records increase.
- Workers are geographically distributed, so the closest or most suitable resource may not be obvious from a flat list.
- Road connectivity is a relationship problem and is more naturally represented as a graph.
- Historical complaint records can contain recurring categories and locations that are difficult to identify without systematic analysis.

4. Problem Understanding and Identification

4.1 Understanding developed during Review 1

Area	Understanding
Existing problem	Large and continuously changing complaint sets can make manual prioritization, assignment and routing slow or inconsistent.
Users affected	Citizens, municipal/utility departments, field workers and supervisors/administrators.
Core information	Complaint category, severity, location, timestamp, status, department, worker availability and road/network information.
Core operations	Insert, search, prioritize, assign, route, update status and analyze historical records.
Main DSA need	Priority-based processing, efficient indexed access and graph-based location/resource relationships.
Research direction	Combine the operational strengths observed in the literature with an explicit DSA-focused processing layer.

4.2 Problem formulation

The project treats complaint handling as a pipeline rather than as one single algorithm. First, a complaint is captured and represented as structured data. Next, its category and priority-related attributes are available to the processing layer. Pending complaints are ordered according to priority. A suitable department and field worker are then considered using availability and location. The road network is represented as a graph, allowing a shortest-path method to estimate an efficient route. Finally, resolution and historical information can be retained for later analysis.

```

Complaint Submission
  ↓
Complaint Record + Category + Severity + Location
  ↓
Priority Calculation
  ↓
Max Heap / Priority Queue
  ↓
Department + Worker Selection
  ↓
Graph Representation of Service Area
  ↓
Shortest-Path Route Selection
  ↓
Resolution + Status Update
  ↓
Historical Frequency / Hotspot Analysis

```

5. Existing Public Utility Complaint Management and Research Background

During Review 1, existing research was examined to avoid presenting the project as if digital complaint reporting itself were new. The literature indicates that modern systems increasingly combine citizen reporting with AI classification, geolocation, routing, dashboards, predictive analytics and field-service workflows.

Aspect	Observed research direction	Project focus
Complaint capture	Digital submission through web/mobile interfaces, often with text, images and location.	Structured complaint record suitable for algorithmic processing.
Classification	NLP/ML or image-based methods classify complaint type or criticality.	Classification can feed the DSA workflow; it is not the central DSA contribution.
Prioritization	Severity/category/location can influence priority.	Explicit priority ordering using a heap-based processing structure.
Resource assignment	Some systems route complaints to departments or field personnel.	Location-aware selection of suitable workers/resources.
Location services	Geotagging and map visualization are common.	Model service locations and road connectivity as a graph for operational routing.
Analytics	Historical complaints are used for recurring issues and proactive governance.	Frequency/hotspot analysis as a later analytical layer.
Academic contribution	Many systems emphasize AI, platform integration or domain-specific governance.	Transparent DSA design with operation-level complexity and evaluation.

This comparison is used for scope definition. The project does not claim to reproduce or outperform the reviewed platforms; it focuses on a transparent algorithmic design suitable for a DSA-II academic evaluation.

6. Literature Review — Four Selected Papers

Four recent and directly relevant papers were selected for Review 1. They cover complementary aspects of smart complaint management: end-to-end civic utility workflows, AI-based classification and prioritization, proactive semantic analysis, and Indian Urban Local Body water-governance operations.

6.1 A Cross-Platform Smart City Utility Management System with MobileNetV2-Based Complaint Classification (2026)

Authors: Khushi Madan, Avneet Singh Oberoi and Jatinpreet Singh.

The paper presents a cross-platform mobile system for urban infrastructure complaints. It uses role-based access for citizens, workers and administrators, MobileNetV2 for complaint criticality classification, Firebase for authentication/data and real-time alerts, and Google Maps API for geospatial services. The paper emphasizes connecting reporting with operational stakeholders rather than treating the application as only a citizen reporting tool.

A particularly relevant observation is that earlier mobile and location-based systems often concentrated on reporting or visualization, while practical integration of location information with field operations, assignment and complete complaint lifecycle management remains less developed. The paper therefore supports the importance of a workflow that continues from complaint submission toward operational resolution.

Relevance to our project: The paper motivates location-aware complaint handling, role-based workflow and field-resource integration. Our proposed contribution is narrower: explicitly designing the data structures and algorithms that manage priority, resource assignment and route selection.

6.2 AI Based Smart Civic Complaint Management System (2026)

Authors: Dr. Suganthi C, Anusuya S, Dhanusha V and Janani M.

The paper proposes a civic complaint system using text, optional images and location information. Its workflow applies NLP preprocessing, TF-IDF feature extraction and machine-learning classification using Logistic Regression or Naïve Bayes. It also describes CNN-based image analysis, cosine-similarity duplicate detection, priority assignment and routing, predictive analytics, status updates and feedback.

The paper is especially relevant because it demonstrates that complaint prioritization and routing can be downstream decisions after classification. Its decision-making layer assigns high, medium or low priority and routes complaints to the appropriate department. Historical data are also used to identify recurring issues and support proactive governance.

Relevance to our project: The paper establishes the usefulness of classification, duplicate detection, prioritization and routing. Our work focuses on the DSA processing layer after complaint information is available, particularly priority-based scheduling, efficient retrieval and location-aware resource handling.

6.3 Proactive Complaint Management in Public Sector Informatics Using AI: A Semantic Pattern Recognition Framework (2025)

Authors: Marco Esperança, Diogo Freitas, Pedro V. Paixão, Tomás A. Marcos, Rafael A. Martins and João C. Ferreira.

This open-access Applied Sciences paper studies public-sector complaint analysis using a CRISP-DM workflow. It uses pretrained BERT-based sentence embeddings, semantic clustering with K-Means, classification and anomaly detection to discover patterns in heterogeneous complaint narratives. The study reports a 27% reduction in repeated complaint categories and a 32% increase in classification efficiency in its evaluated setting.

An important technical observation is that semantic processing can group complaints that express similar problems using different wording. The study also considers model transparency, bias and data governance. Its emphasis is proactive analysis rather than direct DSA-based scheduling.

Relevance to our project: The paper supports the historical-analysis side of complaint management. Our proposed system can treat historical complaint patterns as an analytical input while keeping real-time priority processing and field routing as a separate algorithmic layer.

6.4 CiviCare: An AI-Powered Smart Water Governance and Grievance Resolution System for Urban Local Bodies in India (2026)

Authors: Tanishka Ashish Shah, Dhanashri Jamdar Pingale, Purva Hanumant Pawar, Sakshi Santosh Tupe and Sweetey Mohite.

CiviCare is tailored to Indian Urban Local Bodies and provides a full-stack water-governance workflow. Its grievance module includes NLP-based classification and priority scoring, automated work-order creation, assignment to field plumbers and SLA tracking. The platform also provides ward-level supply dashboards, SMS communication and an AI chatbot.

The paper reports testing on a controlled dataset and evaluates complaint classification and priority scoring. Its rule-based NLP module achieved 88% complaint-type accuracy and 92% priority-assignment accuracy on its stated test set. The authors also identify multilingual and vague descriptions as areas for future improvement.

Relevance to our project: CiviCare is important for the Indian municipal context and shows how complaint prioritization can be connected to field assignment and SLA tracking. Our project generalizes beyond a single utility and concentrates on the DSA mechanisms required to process the operational queue and location network efficiently.

6.5 Paper-to-project comparison

Paper	Main focus	What we take from it
Madan et al. (2026)	Cross-platform smart-city utility workflow, MobileNetV2 criticality, geolocation and roles.	End-to-end lifecycle, field operations and location-aware service delivery.
Suganthi et al. (2026)	NLP/ML classification, image analysis, duplicate detection, prioritization, routing and analytics.	Complaint data can be transformed into decisions for priority and routing.
Esperança et al. (2025)	BERT embeddings, semantic clustering, classification and proactive pattern recognition.	Historical complaints can support recurring-pattern and hotspot analysis.
Shah et al. (2026)	Indian ULB water governance, NLP prioritization, work orders, assignment and SLA tracking.	Indian municipal context and connection between prioritization and field operations.

Overall literature observation: intelligent complaint classification, prioritization, routing and analytics are established research directions. The proposed project therefore positions its academic contribution around the explicit DSA optimization of the operational complaint-processing layer.

7. Research Gaps Identified

The gaps below are stated carefully. A paper's scope is not automatically treated as an author-stated limitation. Where an observation comes from the literature, it is identified as such. Where the project proposes an extension based on the combined scope of the reviewed papers, it is explicitly treated as the project's direction.

Gap / Observation	Basis	How the project responds
Many systems emphasize AI classification or reporting rather than explicit DSA-based operational processing.	Comparison across the four reviewed papers.	Make priority scheduling, retrieval, resource assignment and routing the central algorithmic layer.
Location services are often discussed as geotagging/visualization or as part of routing.	Madan et al. and related-work observations.	Model service locations and road connectivity as a graph and use shortest-path processing for field routing.
Prioritization is present in smart complaint systems but the underlying data structure is not the main research focus.	Suganthi et al. and CiviCare.	Use a Max Heap as the explicit priority-processing structure.
Historical complaint analysis supports proactive governance, but it is separate from	Esperança et al.	Connect historical frequency/hotspot analysis with the operational workflow as a separate analytical layer.

real-time queue processing.		
Systems are often utility-specific or domain-specific.	CiviCare focuses on water governance.	Design the data model for multiple utilities such as roads, water, drainage, waste and streetlights.
Fast operational access becomes important as complaint and worker records grow.	Derived from project requirements, not a direct paper claim.	Study ordered-tree based indexing and structured record access.

7.1 Core project gap statement

Existing research already demonstrates intelligent complaint classification, prioritization, routing and analytics. The proposed project focuses on the underlying algorithmic management problem: how to efficiently prioritize a stream of public utility complaints, organize and retrieve records, connect complaints with suitable service resources, and compute location-aware routes using fundamental Data Structures and Algorithms.

7.2 Important viva clarification

The reviewed papers motivate the complaint-management problem and demonstrate different intelligent modules. They do not prescribe the exact tree, heap or graph structures selected for this project. The DSA structures are project design decisions derived from the required operations. This distinction keeps the literature review genuine and avoids claiming that a paper implemented an algorithm that it did not actually use.

8. Project Objectives

1. Understand the public utility complaint-management problem and its operational challenges.
2. Study recent research on civic complaint classification, prioritization, routing and proactive analysis.
3. Identify a project-specific gap without claiming that existing smart complaint systems do not exist.
4. Study DSA-II Unit 1 and Unit 2 concepts and determine which syllabus topics genuinely fit the problem.
5. Explore priority-based complaint processing using a Max Heap.
6. Explore BST and AVL concepts for ordered/indexed access to complaint or service records.
7. Model service locations, complaint locations and road connectivity using Graph and Adjacency List concepts.
8. Use shortest-path routing as an algorithmic operation over the graph for location-aware field assignment.
9. Explore Tree Traversal for systematic processing and verification of tree-based records.
10. Keep Min Heap, Threaded Binary Tree, Adjacency Matrix, Single-Threaded and Multithreaded concepts appropriately limited or future-oriented where they do not form the core requirement.
11. Prepare the conceptual and algorithmic foundation required for implementation, testing and performance evaluation in later reviews.

9. Stakeholders and User Requirements

Stakeholder	Need	Expected system support
Citizens	Submit and monitor public utility complaints.	Complaint submission, location/details, complaint ID and status tracking.
Municipal/Utility Departments	Receive organized work and manage service workload.	Categorized/prioritized complaint queue, department filtering and workload view.
Field Workers	Know which complaint to handle and how to reach it.	Assigned work item, location information and route guidance.
Supervisors/Administrators	Monitor response performance and resource utilization.	Priority queue view, worker availability, status monitoring and hotspot summaries.
System Administrators	Maintain system records and service configuration.	Complaint, worker, department, location and road-network management.

9.1 Citizen-side requirement flow

1. Citizen provides complaint description and selects/provides location.
2. System stores the complaint as a structured record.
3. Complaint category and priority-related attributes are made available.
4. Complaint enters the priority-processing structure.
5. Department and field resource are selected.
6. Citizen can receive status updates as the work progresses.

10. Functional and Data Requirements

10.1 Functional requirements

- Accept public utility complaints with category, description and location.
- Generate a unique complaint identifier and record submission time.
- Maintain complaint priority information using severity and other defined factors.
- Insert and process pending complaints according to priority.
- Retrieve complaint, department and worker information efficiently.
- Identify a responsible utility department or service area.
- Select an available field worker using location and availability information.
- Represent roads/service locations as a graph and determine a suitable route.
- Update complaint status from pending to assigned, in-progress and resolved.
- Retain historical records for frequency and hotspot analysis.
- Provide measurable performance observations during later implementation.

10.2 Proposed complaint record

```
Complaint {  
  id,  
  category,  
  description,  
  severity,  
  priorityScore,  
  location,  
  submittedAt,  
  status,  
  departmentId,  
  assignedWorkerId  
}
```

10.3 Proposed worker record

```
Worker {  
  id,  
  departmentId,  
  currentLocation,  
  availability,  
  activeJobs,  
  serviceArea  
}
```

10.4 Proposed location/road record

```
RoadEdge {  
  sourceNode,  
  destinationNode,  
  distanceOrTravelCost,  
  roadStatus  
}
```

10.5 Why these attributes were selected

Attribute	Purpose
Severity	Represents urgency or potential service impact.
Category	Helps determine responsible utility department and service type.

Location	Connects the complaint to service areas, workers and road network.
Submission time	Supports waiting-time contribution and fairness in prioritization.
Status	Separates pending, assigned, active and resolved work.
Worker availability	Prevents assigning work to unavailable resources.
Road connectivity	Supports graph-based route selection.
History	Supports recurring issue and hotspot analysis.

11. DSA-II Concepts Studied and Applicability Analysis

The DSA analysis follows the same 12 syllabus topics used in the Review 1 progress report. Each topic is evaluated against an actual system operation instead of being included merely to satisfy a checklist. Algorithms such as shortest-path routing are treated as operations under the Graph concept rather than as additional DSA-topic rows.

DSA Topic	Best TC	Worst TC	Useful?	Application / Reason
Binary Tree	$O(n)$	$O(n)$	No	Foundation for tree concepts, but lacks an ordering rule for efficient complaint-record search.
BST	$O(\log n)$	$O(n)$	Yes	Can provide ordered indexing/search for complaint, department or worker records.
Min Heap	$O(1)$ get min	$O(\log n)$	Limited	Useful when minimum value has priority, but urgent complaints are naturally handled by maximum priority.
Max Heap	$O(1)$ get max	$O(\log n)$	Yes	Core priority structure for repeatedly selecting the highest-priority pending complaint.
Threaded Binary Tree	$O(n)$ traversal	$O(n)$ traversal	No	Can support stack-free traversal, but this is not a dominant requirement of the system.
AVL Tree	$O(\log n)$	$O(\log n)$	Yes	Balanced ordered index for predictable search, insertion and deletion.
Tree Traversal	$O(n)$	$O(n)$	Support	Used to inspect/process tree records; inorder can produce ordered output.
Graph	$O(V+E)^*$	$O(V+E)^*$	Yes	Represents road networks, service locations and relationships between complaints/resources.
Adjacency List	$O(V+E)^*$	$O(V+E)^*$	Yes	Preferred graph representation for a sparse road/service network.
Adjacency Matrix	$O(1)$ edge lookup	$O(1)$ edge lookup	Not preferred	Direct edge checks are fast, but $O(V^2)$ storage is costly for a sparse network.
Single-Threaded	Depends	Depends	Yes	Initial implementation can be sequential for correctness, debugging and reproducible testing.
Multithreaded	Depends	Depends	Future	Could support concurrent requests or parallel scoring later, after the sequential core is stable.

* For graph traversal, the $O(V+E)$ complexity assumes an adjacency-list representation and standard traversal. The exact operation cost depends on the algorithm and representation.

11.1 DSA selection philosophy

The project has three major algorithmic needs. First, pending complaints must be prioritized so that high-priority work can be selected efficiently. Second, records and relationships must be organized so that complaint, department and worker information can be accessed systematically. Third, locations and roads must be represented as a connected network so that a route can be computed. Based on these operations, Max Heap, ordered trees and Graph/Adjacency List are the strongest core concepts.

11.2 What is deliberately not forced into the core

Binary Tree, Threaded Binary Tree and Adjacency Matrix are retained as studied concepts but are not primary structures because their strongest advantages do not align with the current dominant operations. Min Heap is useful for minimum-oriented queries but is not the main priority direction. Multithreading is kept as a future enhancement because the first milestone should establish correctness with a controlled sequential implementation.

12. Detailed Tree-Based Analysis

12.1 Why trees are considered

Complaint-management systems contain records that may need indexed or ordered access. A tree can organize records around a key such as complaint ID, worker ID, department ID or another normalized field. The choice of tree matters because a normal Binary Tree has no ordering property, while BST and AVL are specifically designed for ordered search.

12.2 Binary Tree — why it is not the primary structure

A Binary Tree allows each node to have at most two children, but it does not require smaller values to be placed on one side and larger values on the other. Therefore, searching for a particular complaint or worker may require visiting many nodes, giving $O(n)$ worst-case search. It remains academically important because it is the foundation for BST and AVL.

12.3 Binary Search Tree — baseline ordered indexing

A BST maintains the rule that keys smaller than a node are stored in its left subtree and larger keys in its right subtree. This makes ordered search possible. If the tree is reasonably balanced, search, insertion and deletion are typically $O(\log n)$; if it becomes skewed, the operations can degrade to $O(n)$.

Possible project use: a complaint index can be keyed by complaint ID, a worker index by worker ID, or a department index by department ID. The exact key should be finalized during implementation based on which queries are most frequent.

```
Insert complaint record
↓
Compare complaint key with current tree node
↓
Smaller → left      Larger → right
↓
Reach empty position → insert node
↓
Search follows the same ordering decisions
```

12.4 AVL Tree — preferred balanced tree

An AVL Tree is a self-balancing BST. Its balance factor is $\text{height}(\text{left subtree}) - \text{height}(\text{right subtree})$, and the allowed values are -1, 0 and +1 after rebalancing. If an insertion or deletion causes imbalance, rotations restore the balance.

AVL operation	Project purpose	Complexity
Search	Find a record using its indexed key	$O(\log n)$ guaranteed
Insert	Add a complaint/resource record to the index	$O(\log n)$ guaranteed
Delete	Remove an obsolete/inactive indexed record	$O(\log n)$ guaranteed
Inorder traversal	Produce indexed records in sorted order	$O(n)$

The reason for retaining both BST and AVL in the analysis is conceptual clarity: BST provides the baseline ordered-tree solution and illustrates skewing, while AVL provides predictable logarithmic performance. A later implementation can compare their behavior experimentally.

12.5 Min Heap — limited use

A Min Heap places the smallest key at the root. It could be useful for queries such as the least waiting time, lowest distance or another minimum-oriented policy. However, the primary complaint objective is to select the highest-priority case, so Min Heap is not the main queue structure.

12.6 Max Heap — core priority structure

A Max Heap places the largest key at the root. If the system computes a complaint `PriorityScore`, the complaint with the highest score can be obtained immediately at the top of the heap. Insertion and extraction require $O(\log n)$, while peek is $O(1)$.

```
Pending complaints
↓
PriorityScore calculation
↓
Insert (PriorityScore, ComplaintID) into Max Heap
↓
Highest score at root
↓
extractMax()
↓
Next urgent complaint
```

12.7 Threaded Binary Tree — rejection rationale

A Threaded Binary Tree uses otherwise-null child pointers to link inorder predecessor or successor nodes, reducing dependence on recursion or an explicit stack during traversal. The present project does not have extremely frequent stack-free inorder traversal as

a dominant requirement, so adding this structure would increase complexity without addressing the main complaint-prioritization problem.

12.8 Tree Traversal — supporting operation

Traversal	Complexity	Potential project role
Inorder	O(n)	Produce complaint/resource records in sorted key order from BST/AVL.
Preorder	O(n)	Structural inspection or export of tree organization.
Postorder	O(n)	Bottom-up processing and cleanup-style operations.
Level-order	O(n)	Breadth-wise inspection of tree height and structure.

13. Detailed Graph-Based Analysis

13.1 Why a graph is necessary

Public utility field operations contain many-to-many spatial relationships. A complaint belongs to a location, a worker can serve multiple locations, several complaints can lie in the same service area, and road segments connect multiple locations. A tree is fundamentally hierarchical, whereas a graph naturally represents arbitrary connectivity.

13.2 Proposed graph model

Complaint Location	—	LOCATED_AT	—>	Location Node
Location Node	—	CONNECTED_TO	—>	Location Node
Worker	—	CURRENT_AT	—>	Location Node
Department	—	SERVES	—>	Service Area
Complaint	—	ASSIGNED_TO	—>	Worker

13.3 Graph components

Component	Representation
Vertex	Road junction, service location, ward/zone or other normalized location node.
Edge	Road or direct connectivity between two location nodes.
Weight	Distance, estimated travel time or another route cost.
Worker position	Current worker location represented by a graph node or mapped location.
Complaint position	Service location associated with a complaint.
Adjacency List	For each graph node, store its directly connected nodes and edge weights.

13.4 Adjacency List vs Adjacency Matrix

Aspect	Adjacency List	Adjacency Matrix
Space	O(V+E)	O(V²)
Best suited for	Sparse graphs	Dense graphs
Road-network fit	Preferred because each location connects to a limited number of nearby roads.	Less preferred because all possible node pairs reserve space.
Edge check	O(degree) in a standard list representation	O(1)
Traversal	O(V+E)	O(V²)

13.5 Shortest-path routing as a Graph operation

Once complaint and worker locations are mapped to graph nodes, a shortest-path method such as Dijkstra can be applied when road weights are non-negative. The purpose is not simply to draw a map but to compute a route cost that can support resource assignment.

For example, if Worker W01 is at node A and a complaint is at node F, the graph may contain several possible paths. Dijkstra evaluates cumulative route cost and identifies the least-cost path from A to F. The resulting distance or travel estimate can become one factor in selecting the most suitable worker.

Dijkstra is therefore treated as a graph-based routing algorithm in this documentation, not as an additional row in the 12-topic DSA applicability table.

Worker location A
↓

```

Road graph + edge weights
↓
Dijkstra / shortest-path processing
↓
Complaint location F
↓
Route + estimated cost
↓
Resource assignment decision

```

14. Proposed Priority and Resource Allocation Method

14.1 Why priority must be explicit

First-come-first-served processing treats all pending complaints similarly even when their consequences are different. A blocked drainage point, major water leakage or dangerous road obstruction may deserve earlier attention than a low-impact service request. The project therefore proposes a numerical priority score that can be used by a Max Heap.

14.2 Proposed priority dimensions

Factor	Example interpretation
Severity	How serious is the reported service issue?
Service impact	How many users/areas may be affected?
Waiting time	How long has the complaint remained unresolved?
Category criticality	Some utility categories may have predefined urgency rules.
Location sensitivity	Critical public locations can receive higher weight where appropriate.

14.3 Conceptual priority formula

```

PriorityScore = w1×Severity
               + w2×ServiceImpact
               + w3×WaitingTime
               + w4×CategoryCriticality
               + w5×LocationSensitivity

```

where the weights are normalized and finalized only after testing.

The weights are intentionally not fixed in Review 1. Choosing numerical values without experiments would create a false impression of validation. During implementation, several weight configurations can be compared using response-time and ranking-related observations.

14.4 Resource assignment concept

After selecting the next complaint, the system can identify workers belonging to the responsible department and filter out unavailable resources. Among feasible workers, route cost and workload can be considered. The graph is used to calculate travel cost; the final assignment rule can combine distance with availability and current workload.

```

Highest-priority complaint
↓
Identify responsible department
↓
Filter available workers
↓
Map worker + complaint locations to graph
↓
Compute route cost
↓
Select feasible worker using defined policy

```

↓
Create assignment

15. Proposed End-to-End System Workflow

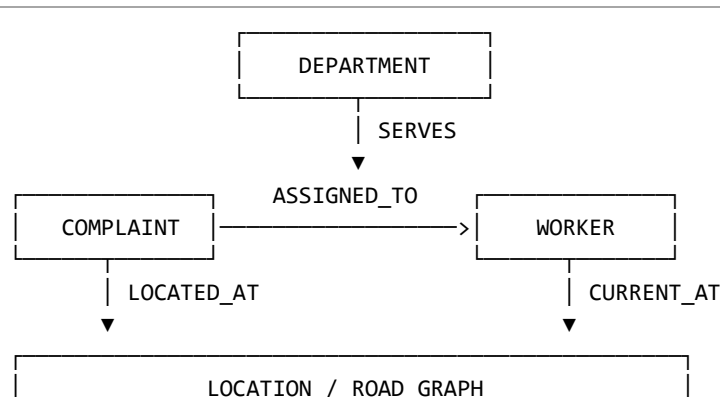
STEP 1 Citizen submits complaint
↓
STEP 2 Store complaint record + location + timestamp
↓
STEP 3 Category / severity information is determined
↓
STEP 4 Calculate PriorityScore
↓
STEP 5 Insert complaint into Max Heap
↓
STEP 6 Extract highest-priority pending complaint
↓
STEP 7 Identify responsible department/service area
↓
STEP 8 Retrieve available field workers
↓
STEP 9 Use Graph + shortest-path processing for route cost
↓
STEP 10 Select and assign suitable worker
↓
STEP 11 Worker updates status and resolution
↓
STEP 12 Store historical outcome for analytics

15.1 Why the stages are separated

Stage	Reason for separation
Complaint capture	Creates a consistent structured input.
Priority calculation	Converts urgency-related attributes into a comparable value.
Heap processing	Efficiently selects the highest-priority pending case.
Department/worker filtering	Avoids considering clearly unsuitable resources.
Graph routing	Solves the spatial connectivity problem separately from priority.
Resolution tracking	Maintains service lifecycle state.
Historical analysis	Separates long-term trends from immediate operational decisions.

16. Conceptual Data Model

16.1 Relationship model



Node — CONNECTED_TO(weight) — Node ...

16.2 Example complaint records

Complaint	Category	Severity	Location	Status
C101	Road Damage	High	Node N14	Pending
C102	Streetlight	Medium	Node N21	Pending
C103	Water Leakage	High	Node N08	Assigned
C104	Garbage	Low	Node N17	Pending

16.3 Example worker records

Worker	Department	Location	Availability	Active Jobs
W01	Roads	N11	Available	1
W02	Water	N05	Available	0
W03	Sanitation	N19	Busy	2

These records are illustrative only. They are not claimed as real municipal data and are intended to demonstrate how the data structures will interact during implementation.

17. Proposed Algorithms and Pseudocode

17.1 Priority calculation

```
calculatePriority(complaint):
    severityScore = normalize(complaint.severity)
    impactScore   = normalize(complaint.serviceImpact)
    waitingScore  = normalize(waitingTime(complaint))
    categoryScore = categoryCriticality(complaint.category)
    locationScore = locationSensitivity(complaint.location)
    return w1*severityScore + w2*impactScore
        + w3*waitingScore + w4*categoryScore
        + w5*locationScore
```

17.2 Max Heap complaint processing

```
for each incoming complaint:
    score = calculatePriority(complaint)
    maxHeap.push(score, complaint.id)

while maxHeap is not empty:
    nextComplaint = maxHeap.extractMax()
    process(nextComplaint)
```

17.3 Ordered record search using BST/AVL

```
search(root, key):
    if root == NULL:
        return NOT_FOUND
    if key == root.key:
        return root.record
    if key < root.key:
        return search(root.left, key)
    else:
        return search(root.right, key)
```

In the final implementation, AVL can replace the unbalanced BST while preserving the same search interface. The difference is that AVL maintains balance after insertion and deletion.

17.4 Graph traversal / route preparation

```
routeCost(source, destination):
    initialize distance[source] = 0
    initialize remaining distances = infinity
    repeatedly select the unprocessed node
    relax outgoing edges with smaller cumulative cost
    stop when destination is finalized
    return shortest distance / predecessor path
```

The above is a conceptual shortest-path sequence corresponding to Dijkstra-style processing for non-negative edge weights. Exact implementation details will be finalized during coding.

17.5 Resource selection

```
selectWorker(complaint):
    department = findDepartment(complaint.category)
    candidates = availableWorkers(department)
    for each worker in candidates:
        route = shortestPath(worker.location, complaint.location)
        evaluate route cost + workload policy
    return best feasible worker
```

18. Complexity and Performance Considerations

Component	Typical complexity	Why it matters
BST search	$O(\log n)$ average; $O(n)$ worst	Efficient ordered access but can degrade when skewed.
AVL search/insert/delete	$O(\log n)$ guaranteed	Predictable performance as indexed records grow.
Tree traversal	$O(n)$	Each relevant node may need to be processed.
Max Heap peek	$O(1)$	Immediate access to the highest-priority complaint.
Max Heap insert	$O(\log n)$	Maintains heap ordering after a new complaint arrives.
Max Heap extractMax	$O(\log n)$	Retrieves the next highest-priority complaint.
Graph traversal with adjacency list	$O(V+E)$	Efficient for sparse road/service networks.
Adjacency List storage	$O(V+E)$	Stores actual connections instead of all possible pairs.
Adjacency Matrix storage	$O(V^2)$	Can become expensive as the location network grows.
Dijkstra with suitable implementation	Depends on representation/priority structure	Used to compute route cost over the weighted graph.

18.1 Evaluation planned for later reviews

- Compare BST and AVL search behavior on increasing record counts.
- Measure Max Heap insertion and extraction behavior as pending complaints increase.
- Measure graph traversal and route computation as vertices and edges increase.
- Compare adjacency-list and adjacency-matrix space behavior on sample graph sizes.
- Measure complaint-processing time from insertion to selection.
- Compare different priority-weight configurations after a suitable test dataset is available.
- Evaluate assignment consistency and route cost on sample worker/complaint locations.
- Document correctness through sample inputs, expected outputs and test cases.

18.2 Important limitation of Review 1

No final accuracy, speed, route-quality or response-time result is claimed in Review 1 because the integrated implementation and evaluation dataset are not yet available. Complexity statements describe the expected algorithmic behavior, not measured application performance.

19. Work Completed During Month 1

- Finalized the project title and public-utility complaint-management direction.
- Defined the problem in terms of priority processing, resource assignment and location-aware routing.
- Identified target stakeholders and the information that must be managed.
- Reviewed exactly four selected research papers relevant to smart complaint management.
- Compared the papers and separated literature observations from project-specific design decisions.
- Identified the project-specific research gap around explicit DSA-based operational optimization.
- Studied DSA-II Unit 1 and Unit 2 concepts relevant to the project.
- Evaluated the 12 DSA topics used in the Review 1 applicability table.
- Designed the conceptual complaint-priority and Max Heap workflow.
- Designed the location/road graph and adjacency-list representation.
- Mapped worker assignment and shortest-path routing to graph operations.
- Prepared the initial data model, pseudocode and complexity direction.
- Prepared the implementation and evaluation plan for Review 2.

19.1 What has NOT been claimed as completed

- The complete complaint-management application has not yet been implemented.
- The final database and production deployment have not yet been completed.
- The final priority weights have not been experimentally validated.
- The final worker-assignment policy has not yet been benchmarked.
- No real municipal operational dataset has been used for final performance evaluation.
- No final response-time or routing-performance result is claimed for the proposed system.

20. Evidence and Supporting Material

The concise 3-page PBL Progress Report – I is the formal Review 1 summary. This detailed document is intended to sit behind that report as supporting evidence and to explain how the reported items were reached.

Evidence item	What it demonstrates
Problem statement and context	Understanding of the public utility complaint problem and its operational difficulty.
Four-paper literature review	Research activity and basis for the proposed project direction.
Research-gap table	Connection between literature observations and the project's DSA-focused scope.
DSA applicability table	Reasoned selection, limitation and rejection of the 12 syllabus concepts.
Priority workflow	Understanding of Max Heap based complaint scheduling.
Graph model	Understanding of complaint locations, workers, departments and road relationships.
Pseudocode	Initial understanding of how selected operations will be implemented.
Complexity table	Awareness of computational cost and scalability.
Review 2 plan	Transition from conceptual analysis to implementation and testing.

20.1 Evidence placement

For submission, the concise progress report can remain the official summary, while this detailed documentation can be attached or shown as the supporting evidence document. As implementation progresses, screenshots of source code, console output, graph visualizations, sample complaint queues and test results should be added.

21. Challenges Faced and How They Were Addressed

Challenge	Why it was difficult	How it was addressed
Relating DSA to a real public-service problem	It is easy to describe a complaint website but harder to identify genuine algorithmic operations.	Mapped each selected concept to priority processing, indexed access, traversal or relationship/routing operations.
Avoiding forced DSA usage	Not every syllabus structure solves a major project requirement.	Evaluated all 12 topics and documented Yes/No/Limited/Future decisions.
BST vs AVL	Both support ordered search and may appear redundant.	BST is treated as the baseline; AVL is the balanced option for predictable performance.
Tree vs Graph	Both organize data but represent fundamentally different structures.	Trees handle ordered/indexed records; graphs handle spatial and many-to-many relationships.
Min Heap vs Max Heap	Both are priority structures but prioritize opposite directions.	Max Heap is selected for highest-priority complaints; Min Heap remains limited for minimum-oriented queries.

Turning location into an algorithmic model	A map alone does not demonstrate the DSA operation.	Roads and service locations are represented as weighted graph nodes and edges.
Understanding research gaps	A paper's scope is not automatically a stated limitation.	Separated author-supported observations from project-specific extensions.
AI vs DSA boundary	Smart complaint papers often contain substantial AI components.	AI/NLP is treated as an optional upstream classifier; the academic core remains DSA-based operational processing.
Designing without overcomplication	Adding every possible feature could make the project unrealistic for a DSA course.	Kept Review 1 centered on priority, ordered records, graph modelling and routing.

22. Individual Contribution

As this is an Individual Assignment, all work documented in Review 1 was carried out by me. My contribution includes both research and technical analysis.

- Defined the public utility complaint-management problem and project scope.
- Studied the operational difficulty of prioritizing, assigning and routing many complaints.
- Studied recent smart complaint-management approaches.
- Read and analyzed the four selected research papers.
- Identified research observations and formulated the project-specific gap.
- Studied DSA-II Unit 1 and Unit 2 concepts relevant to the project.
- Compared DSA structures using operation complexity and project suitability.
- Designed the complaint-priority and Max Heap processing concept.
- Designed the location and road graph model.
- Designed the initial location-aware resource-assignment workflow.
- Prepared the initial pseudocode and complexity analysis.
- Prepared the Review 1 report and this detailed supporting documentation.

23. Current Status and Progress

The reported overall progress is 25%. This percentage represents completion of the analytical foundation rather than 25% of a fully implemented software product.

Work area	Status	Evidence in this documentation
Problem understanding	Completed	Sections 3–4
Stakeholder/requirements analysis	Completed	Sections 9–10
Existing-system/research background	Completed at conceptual level	Section 5
Four-paper literature review	Completed	Section 6
Research-gap identification	Completed	Section 7
DSA-II Trees study	Completed	Sections 11–12
DSA-II Graphs study	Completed	Sections 11 and 13
Initial DSA mapping	Completed	Sections 11–18
Priority/routing workflow	Completed conceptually	Sections 14–17
Implementation	Not yet completed	Planned for Review 2
Testing/performance evaluation	Not yet completed	Planned for later reviews

24. Plan for Review 2

The next review moves from conceptual analysis toward implementation. The focus will remain on correctly applying the selected DSA concepts rather than adding unnecessary features.

1. Study the relevant DSA-II topics from the next units and understand their operations and implementation.
2. Finalize the complaint, department, worker and location data representation.
3. Create a small sample dataset of complaints, workers, departments and road nodes.
4. Implement the baseline BST and compare it conceptually/experimentally with AVL.
5. Implement Max Heap insertion, peek and extractMax operations for complaint priority.
6. Implement the graph using an adjacency list.
7. Implement shortest-path processing for worker-to-complaint route calculation.
8. Integrate priority processing with department and worker selection.
9. Create sample test cases covering high-, medium- and low-priority complaints.

10. Test insertion, search, traversal, heap operations and routing on increasing sample sizes.
11. Collect evidence: source-code screenshots, diagrams, sample inputs/outputs and complexity observations.
12. Document Review 2 results and compare expected versus observed behavior.

24.1 Review 2 implementation target

```
Sample Complaint
↓
Category + Severity + Location
↓
PriorityScore
↓
Max Heap
↓
Highest-Priority Complaint
↓
Department + Available Workers
↓
Graph / Adjacency List
↓
Shortest-Path Route
↓
Assigned Worker + Test Output
↓
Complexity / Performance Observation
```

25. Conclusion

Review 1 established the conceptual and algorithmic foundation of the Smart Public Utility Complaint Management System. The project began by recognizing that modern civic complaint platforms already provide reporting, classification, geolocation, routing, tracking and analytics. The proposed work therefore does not treat the problem as simply “build a complaint website.” Instead, it focuses on the underlying algorithmic challenge of efficiently organizing and processing a growing stream of public utility complaints.

The four selected papers provide complementary perspectives. Madan et al. demonstrate a cross-platform smart-city workflow connecting citizens, administrators and workers with criticality classification and geospatial services. Suganthi et al. demonstrate NLP/ML classification, duplicate detection, prioritization, routing and predictive analytics. Esperança et al. demonstrate semantic pattern recognition and proactive complaint analysis. Shah et al. demonstrate an Indian ULB-oriented water governance platform connecting complaint prioritization with field work orders and SLA tracking.

From these observations, the project identifies a defensible gap: the explicit DSA-based optimization of the operational complaint-processing layer. The proposed system uses a Max Heap for priority-based complaint selection, BST/AVL concepts for ordered record access, Graph and Adjacency List for location/road relationships, and shortest-path processing for location-aware worker routing. Tree traversal supports systematic inspection, while Min Heap, Threaded Binary Tree, Adjacency Matrix and multithreading are intentionally treated as limited, alternative or future concepts where appropriate.

The project is intentionally not presented as implemented at this stage. Review 1 establishes the problem, literature foundation, DSA mapping, conceptual data model, workflow, pseudocode and evaluation direction. The next step is to convert this design into a small working implementation, validate the selected operations on sample data and collect measurable evidence for later reviews.

26. References

- [1] Madan, K., Oberoi, A. S., & Singh, J. (2026). A Cross-Platform Smart City Utility Management System with MobileNetV2-Based Complaint Classification. *International Journal of Advances in Science Transactions — Soft Computing and Intelligent Systems (IJASCIS)*, 5(1), 22–39.
- [2] Suganthi, C., Anusuya, S., Dhanusha, V., & Janani, M. (2026). AI Based Smart Civic Complaint Management System. *International Journal of Engineering & Extended Technologies Research*, 8(2). DOI: 10.15662/IJEETR.2026.0802433.
- [3] Esperança, M., Freitas, D., Paixão, P. V., Marcos, T. A., Martins, R. A., & Ferreira, J. C. (2025). Proactive Complaint Management in Public Sector Informatics Using AI: A Semantic Pattern Recognition Framework. *Applied Sciences*, 15, 6673. DOI: 10.3390/app15126673.

- [4] Shah, T. A., Pingale, D. J., Pawar, P. H., Tupe, S. S., & Mohite, S. (2026). CiviCare: An AI-Powered Smart Water Governance and Grievance Resolution System for Urban Local Bodies in India. *Indian Journal of Modern Research and Reviews*, 4(5), 318–324. DOI: 10.5281/zenodo.20429097.
- [5] Smart Public Utility Complaint Management System — PBL Progress Report I, Review 1, Data Structure and Algorithms - II (CCSE0301), prepared for the individual assignment.
- [6] Intelligent Freelance Marketplace Platform — Review 1 Detailed Documentation, supplied as the reference document for structure and level of detail.

Source discipline note: The four research papers are used to motivate the literature review and research-gap discussion. The exact DSA choices are project design decisions derived from the operations required by the proposed system; the reviewed papers are not presented as having prescribed these specific DSA structures.